UED IA-2 PROJECT REPORT

Campus Navigation App

FOR ACCESSIBILITY AND INCLUSIVITY

Course: User Experience Design (UED)

Semester: LY B.Tech (July-Nov 2025)

Project: UED IA-2 - Campus Navigation App

TEAM MEMBERS

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1. Problem Definition

Navigating large university campuses like **Somaiya Vidyavihar University** can be challenging for students, staff, and visitors—especially for those with **mobility or visual impairments**. Current navigation systems and campus apps are not designed with accessibility as a core focus.

The problem identified is the **lack of an inclusive, accessible, and reliable mobile navigation system** that supports all individuals, regardless of physical or sensory ability, to move independently and confidently within the campus.

2. Objectives

The primary objectives of this project are:

1. To design a **mobile navigation app** that ensures **universal accessibility and inclusivity**.
2. To provide **audio-based and visual navigation guidance** for users with varying abilities.
3. To integrate **real-time GPS-based campus wayfinding** that highlights accessible routes.
4. To enhance **user independence and comfort**, reducing anxiety related to campus navigation.
5. To align the application with **global accessibility standards** and UX design best practices.

3. Tools and Technologies Used

- **Design Tool**: Figma, Adobe XD
- **Prototyping Tools**: Figma, Stitch
- **Research Sources**: Evelity, AndAcademy

4. Accessibility and Inclusivity Features

a. Audio Navigation Support

- Step-by-step **voice guidance** compatible with **VoiceOver** and **TalkBack**.
- Enables visually impaired users to move around campus safely and independently.

b. Visual Accessibility

- **High contrast themes**, **large touch targets**, and **minimal text clutter**.
- Suitable for users with low vision or color blindness.

c. Mobility-Friendly Routing

- Routes are **optimized for wheelchair users**, avoiding stairs and steep inclines.
- Highlights **ramps, elevators, and accessible restrooms**.

d. Multilingual Interface

- Supports multiple languages to accommodate **foreign students** and visitors.
- Default English + options for regional and international languages.

e. Real-Time Wayfinding

- Combines **GPS and indoor navigation** for precise location guidance.
- Offers **voice feedback** to alert users at key decision points.

f. Accessibility Information Integration

- Maps display **accessible restrooms, ramps, lifts, and assistance centers**.
- Enables users to make quick and informed decisions.

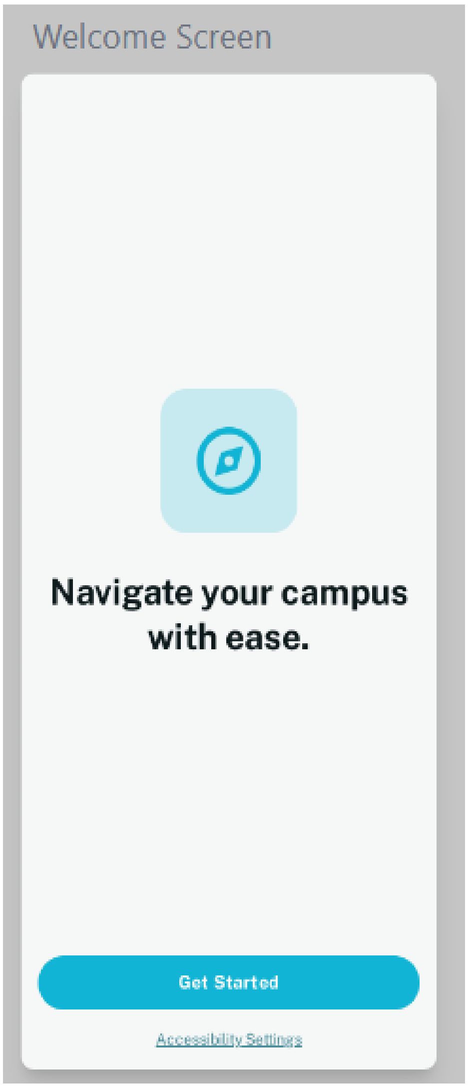
5. Design Principles and Justifications (UX-Focused)

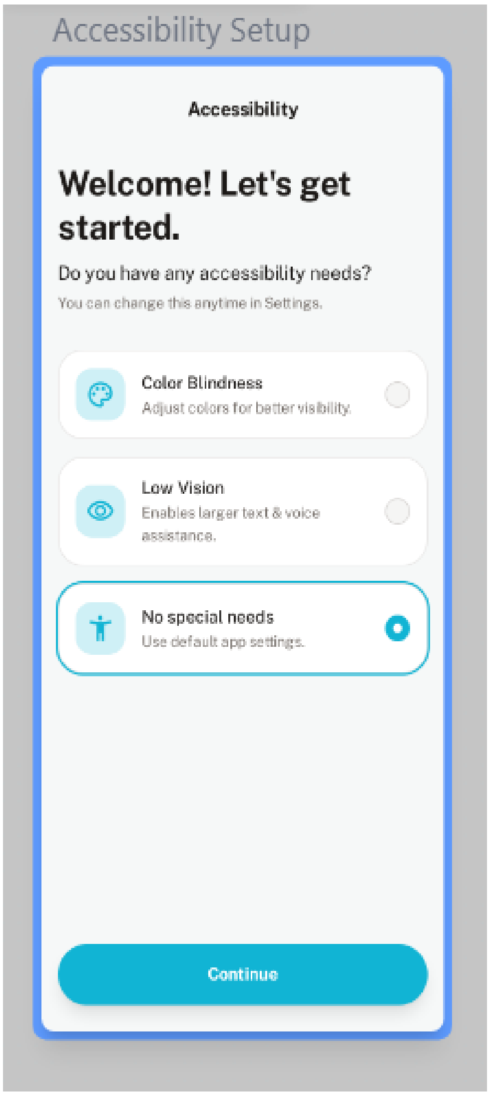
Empathy	User research focused on people with mobility, visual, and auditory disabilities to understand their needs.
Inclusivity	Designed for all users—visually impaired, mobility-challenged, and neurodiverse individuals.
Perceivability	Content accessible through **text, sound, and vibration cues** for multi-sensory communication.
Operability	Works with touch, gestures, or voice commands for users with limited motor control.
Understandability	Clear and simple language; intuitive icons and color coding for reduced cognitive load.
Robustness	Compatible with various devices and assistive technologies ensuring reliable performance.

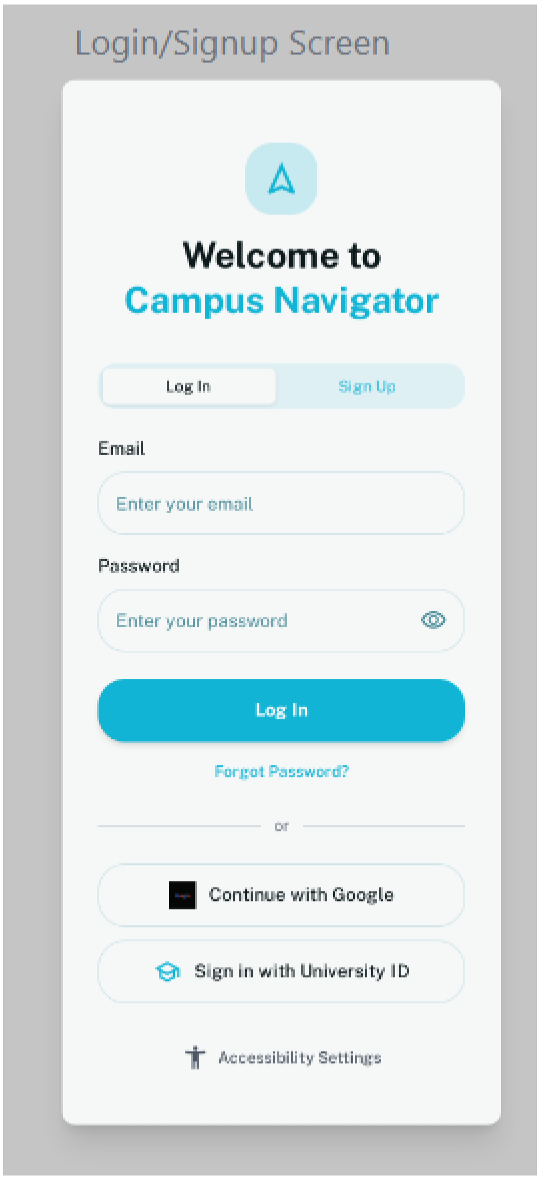
6. Functional Prototype

A functional **interactive prototype** was developed in **Figma** and enhanced in **Stitch**.
The prototype includes:

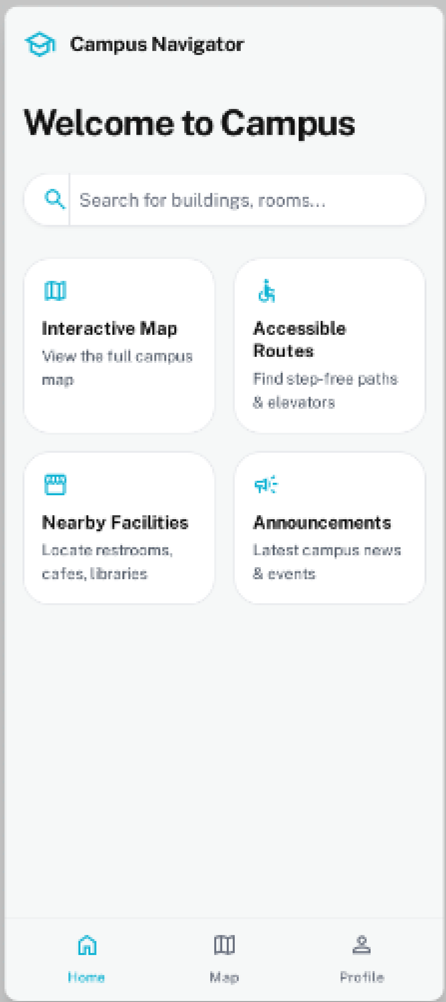
- Home screen with quick-access categories (e.g., Classrooms, Cafeteria, Labs).
- Voice navigation option toggle.
- Route selection screen with accessible path highlights.
- AR-based indoor navigation (mocked for prototype).
- Multi-language dropdown and settings panel.





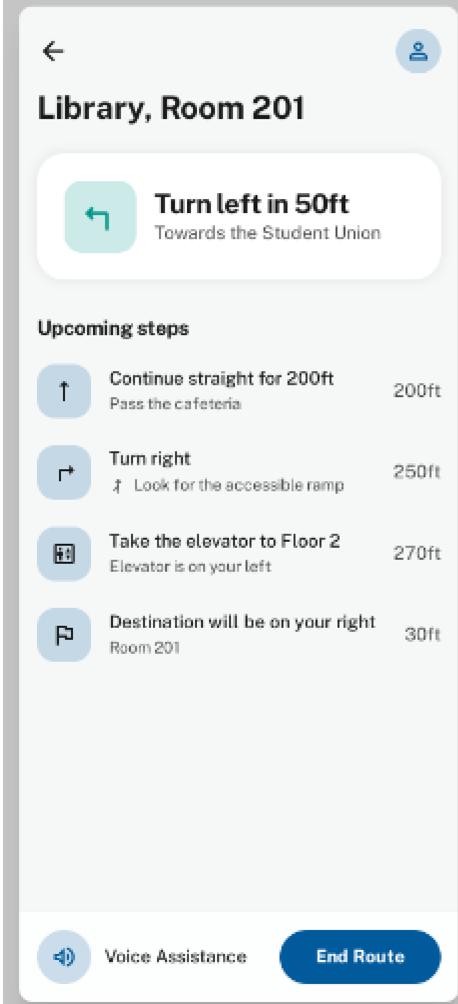


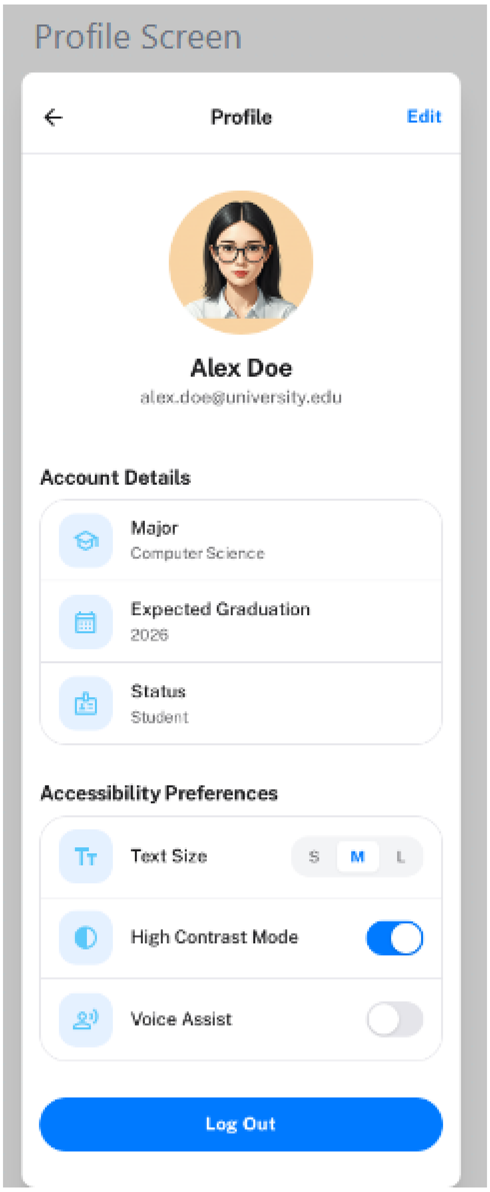
Home Screen



Map Screen

Navigation Screen





7. Results and Findings

- The prototype was **tested with diverse users**, including those with visual impairments and mobility issues.
- Users appreciated **audio cues, high contrast UI**, and **accessibility information visibility**.
- Feedback indicated that the app **reduces stress** during navigation and encourages **independent mobility**.
- The design successfully demonstrates how **accessibility can enhance usability** for

all.

8. Conclusion

The **Campus Navigation App** showcases how user-centered design, when grounded in accessibility and inclusivity, can transform the way people interact with their environment. It not only assists individuals with disabilities but also improves overall campus experience for everyone.

The project aligns with **Somaiya Vidyavihar University's commitment to inclusivity**, adhering to **global UX accessibility standards** and demonstrating practical, real-world impact through thoughtful design.

References

1. Arounda: Accessibility in UX Best Practices
2. Evelity: University Campuses Wayfinding for All
3. Ramotion: Accessibility in UX Design
4. InclusiveCityMaker: Apps for Visually Impaired Users
5. AndAcademy: Accessibility in UX Design
6. Somaiya University Official Site
7. Somaiya App – Play Store (EN)